



UTLA MUKALINGAM

AI Solution Architect

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CORE EXPERTISE

Agentic AI: Multi-Agent Systems, LangGraph, CrewAI, LangChain, Llamaindex, MCP, A2A Protocol, Tool-Calling, Stateful Orchestration, HITL Design

LLM & RAG: Claude, GPT-4o, Gemini, Llama; RAG Pipelines, Hybrid Retrieval, Re-ranking, Semantic Chunking, Fine-tuning (LoRA/QLoRA), Prompt Engineering

LLMOps & Eval: Langfuse, LangSmith, RAGAS, Eval Harnesses, Tracing, Drift Detection, MLflow

Voice AI: ElevenLabs, Whisper, Deepgram, Twilio, LiveKit, Real-time TTS/STT, Sub-800ms Latency

Vector DBs: pgvector, ChromaDB, FAISS, Pinecone, Weaviate

ML / CV: PyTorch, TensorFlow, Scikit-learn, Transformers, LSTM, XGBoost, YOLO, OpenCV

Cloud / DevOps: AWS (EC2, Lambda, SageMaker, S3), Azure (AKS, OpenAI), Docker, Kubernetes, CI/CD, n8n

Backend: FastAPI, Django, Node.js, REST; Next.js 14, React, Tailwind

Data: PostgreSQL, Redis, MySQL, Pandas, NumPy

EDUCATION

B.Tech, Mechanical Engineering
St. Martin's Engineering College · 2015–2019

Prof. Certification — Data Science & AI
360DigiTMG · 2021

PROFESSIONAL SUMMARY

Senior AI Solution Architect with 7+ years of production-grade experience designing and shipping LLM-first, agentic AI systems at enterprise and national scale. Specialist in stateful multi-agent orchestration (LangGraph, CrewAI), deterministic tool-calling, MCP/A2A interoperability, RAG pipelines, and Voice AI — backed by rigorous LLMOps and evaluation. Proven delivery across FinTech, Healthcare, Government, AdTech, and Manufacturing, including a live B2B Voice AI SaaS and government-recognized deployments. Open to relocation (UAE / GCC).

PROFESSIONAL EXPERIENCE

AI Solution Architect

Jan 2026 – Present

IICL — Intelligence India.Com Limited

Hyderabad, India

- › **Architected stateful multi-agent systems** using LangGraph & CrewAI with human-in-the-loop gates and deterministic tool-calling, powering procurement and document-intelligence workflows for enterprise clients.
- › **Engineered end-to-end RAG pipelines** over structured/unstructured data on pgvector with advanced chunking, lifting retrieval precision by 40% and sharply cutting hallucination rates.
- › **Designed AI lead-qualification & appointment automation** integrating Claude API tool-use, RAG, and webhook-driven CRM pipelines — reducing manual sales effort by 50%.
- › **Led AI agent strategy proposals** across Merch, Pricing, and Inventory verticals for global clients — full architecture design through production deployment.
- › **Built iVaak AI** (live at ivaak.ai) — a multi-tenant Voice AI platform (ElevenLabs + LangGraph + Twilio) with isolated per-tenant state machines and real-time human escalation.
- › **Shipped TruFix AI** — an autonomous ITSM SaaS on LangGraph + FastAPI for ticket triage, root-cause analysis, and resolution (~35% MTTR), with full Langfuse LLM observability.

Lead AI Engineer

Apr 2025 – Jan 2026

LowCode Labs Pvt Ltd

Hyderabad, India

- › **Deployed enterprise AI agents** (RAG + GenAI) serving 500+ concurrent users at 99.5% uptime via robust MLOps and CI/CD.
- › **Architected multi-agent workflows** on LangChain + FastAPI, cutting deployment time 60% through automated model versioning and container orchestration on AWS.
- › **Built a procurement intake chatbot** and document-processing automation integrating Claude Vision API and OCR pipelines.
- › **Developed custom React widgets** and low-code AI integrations (PDF annotators, KPI cards, document signers) embedded into enterprise Mendix apps for pharma and FMCG.

Lead AI Engineer

Nov 2023 – Mar 2025

The Audience Street Inc.

Hyderabad, India

- › **Architected an LLM + NLP AdTech platform** achieving a 25% lift in ad relevance and engagement through semantic content optimization.
- › **Integrated GenAI pipelines** for automated ad-copy generation, audience segmentation, and real-time feedback loops — accelerating campaign cycles by 40%.
- › **Built prompt-evaluation harnesses** for cost, latency, and regression benchmarking; cut LLM API spend 30% via intelligent caching and batching. Led a cross-functional team of 6.

CERTIFICATIONS

- IBM — Machine Learning with Python
- Mendix Intermediate Certified Developer
- AI & Deep Learning — Professional Course
- Data Visualization with Tableau

RECOGNITION

- Empanelled Faculty — Boston Institute of Analytics (GenAI & Agentic AI)
- Speaker — East Coast Maritime & Logistics Summit 2025
- Medium Writer — LLMs, RAG & Enterprise Agentic AI

LANGUAGES

English — Professional
Hindi · Telugu — Native

Senior Data Scientist

Dec 2021 – Nov 2023

Spotflock Technologies

Hyderabad, India

- › **Deployed a Computer Vision system** (YOLO + OpenCV) for predictive maintenance at Sembcorp — detecting gauge abnormalities in real time, saving 900+ labor hours and \$1M+ in operating costs.
- › **Built a Transformer-based ocean-current model** at 85% accuracy for INCOIS (Ministry of Earth Sciences), with automated ingestion and scheduled retraining.
- › **Delivered electricity load forecasting** (ARIMA + LSTM) for the Haryana State Government, directly supporting public-utility resource planning.

Data Scientist

Jul 2019 – Nov 2021

Innodatatics Inc.

Hyderabad, India

- › **Reduced equipment downtime 30%** with Python predictive-maintenance models across industrial client portfolios; built end-to-end ML pipelines for time-series, NLP, and CV tasks.

FLAGSHIP PROJECTS

iVaak AI — Multi-Tenant Voice AI SaaS (Live)

2025 – Present

LangGraph · ElevenLabs · Whisper · FastAPI · Twilio · Redis · pgvector

- › Production B2B platform giving each tenant an isolated state machine, vector knowledge base, and custom escalation policy at sub-800ms latency; live with active enterprise clients (ivaak.ai).

TruFix AI — Autonomous ITSM Platform

2025 – Present

LangGraph · Claude API · FastAPI · Langfuse · PostgreSQL

- › AI-native layer above Freshdesk/Zendesk; agents own full ticket lifecycle with confidence-threshold human routing — 35% MTTR reduction in pilots, full Langfuse eval before each release.

Internal Copilot Agent — Microsoft (Confidential)

2026

Azure OpenAI · Azure DevOps API · LangGraph · RAG · MS Teams

- › Enterprise Copilot embedded in Teams that creates/manages ADO work items, auto-generates test cases, submits pipeline jobs, and answers from an integrated knowledge base — cutting manual ADO overhead and accelerating sprint planning.

QConnect — Enterprise AI Training Platform (Qatar Energy)

2025

Next.js 14 · Fastify · FastAPI · pgvector · Claude API · LiveKit · DeepFace

- › Video-learning platform with AI-generated quizzes, face-presence monitoring, and avatar responses, built on a polyglot microservice stack.

Pharma Regulatory Document Intelligence — Dr. Reddy's

2025

FastAPI · Claude API · pgvector · LangChain · RAG

- › RAG-based document-intelligence platform for regulatory and compliance documents at Dr. Reddy's; semantic chunking with grounded, citation-backed answer generation to accelerate retrieval across large pharma document sets.

Ocean Current Prediction — INCOIS (Govt. of India)

2022 – 2023

Transformers · Python · AWS · MLOps

- › National-scale spatiotemporal model at 85% accuracy with automated end-to-end MLOps and drift detection for India's national oceanographic service.

Electricity Load Forecasting — Haryana Govt. (HERC)

2022 – 2023

XGBoost · LSTM · ARIMA · Python · Tableau

- › Ensemble forecasting system on Indian Energy Exchange data, deployed for the Haryana Electricity Regulatory Commission to support state-grid resource planning and stability.

Nexlayer AI — Domain-Specific Agentic Platform

2025 – Present

CrewAI · FastAPI · React · pgvector · Claude API

- › Compliance-first agentic AI platform targeting Finance, Government, and Healthcare in India; 13-screen React prototype with full PRD/TRD and wireframes.